JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Functions of One Real Variable (2015–2021)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
80	133 M	240 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	41	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	16	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	23	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q57 • ID: JAM_2021_Q57) NAT +2 M Neg: Nil (0)

Q. 57 The value of

$$\lim_{n\to\infty} \int_0^1 e^{x^2} \sin(nx)\, dx$$

is _____.

Question 2 (JAM 2021 • Q54 • ID: JAM_2021_Q54) NAT +2 M Neg: Nil (0)

Q. 54 The largest positive number a such that

$$\int_0^5 f(x)dx + \int_0^3 f^{-1}(x)dx \geq a$$

for every strictly increasing surjective continuous function $f : [0, \infty) \to [0, \infty)$ is _____.

Question 3 (JAM 2021 • Q53 • ID: JAM_2021_Q53) NAT +2 M Neg: Nil (0)

Q. 53 Consider those continuous functions $f : \mathbb{R} \to \mathbb{R}$ that have the property that given any $x \in \mathbb{R}$,

$$f(x) \in \mathbb{Q} \text{ if and only if } f(x+1) \in \mathbb{R} \setminus \mathbb{Q}.$$

The number of such functions is _____.

Question 4 (JAM 2021 • Q52 • ID: JAM_2021_Q52) NAT +2 M Neg: Nil (0)

Q. 52 The least possible value of k, accurate up to two decimal places, for which the following problem

$$y''(t) + 2y'(t) + ky(t) = 0, t \in \mathbb{R},$$
$$y(0) = 0, y(1) = 0, y(1/2) = 1,$$

has a solution is _____.

Question 5 (JAM 2021 • Q45 • ID: JAM_2021_Q45) NAT +1 M Neg: Nil (0)

Q. 45 Consider the set $A = \{a \in \mathbb{R} : x^2 = a(a+1)(a+2) \text{ has a real root }\}$. The number of connected components of A is _____.

Question 6 (JAM 2021 • Q38 • ID: JAM_2021_Q38) MSQ +2 M Neg: Nil (0)

JAM 2021 MATHEMATICS - MA

Q. 38 Consider the four functions from $\mathbb{R}$ to $\mathbb{R}$:

$$f_1(x) = x^4 + 3x^3 + 7x + 1, \quad f_2(x) = x^3 + 3x^2 + 4x, \quad f_3(x) = \arctan(x)$$

and

$$f_4(x) = \begin{cases} x & \text{if } x \notin \mathbb{Z}, \\ 0 & \text{if } x \in \mathbb{Z}. \end{cases}$$

Which of the following subsets of $\mathbb{R}$ are open?

(A) The range of f_1.

(B) The range of f_2.

(C) The range of f_3.

(D) The range of f_4.

Question 7 (JAM 2021 • Q35 • ID: JAM_2021_Q35) MSQ +2 M Neg: Nil (0)

Q. 35 Let $f : (a, b) \to \mathbb{R}$ be a differentiable function on (a, b). Which of the following statements is/are true?

(A) $f' > 0$ in (a, b) implies that f is increasing in (a, b).

(B) f is increasing in (a, b) implies that $f' > 0$ in (a, b).

(C) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then there exists a $\delta > 0$ such that $f(x) > f(x_0)$ for all $x \in (x_0, x_0 + \delta)$.

(D) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then f is increasing in a neighbourhood of x_0.

Question 8 (JAM 2021 • Q32 • ID: JAM_2021_Q32) MSQ +2 M Neg: Nil (0)

Q. 32 Consider the equation

$$x^{2021} + x^{2020} + \cdots + x - 1 = 0.$$

Then

(A) all real roots are positive.

(B) exactly one real root is positive.

(C) exactly one real root is negative.

(D) no real root is positive.

Question 9 (JAM 2021 • Q31 • ID: JAM_2021_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 Let $f : \mathbb{R} \to \mathbb{R}$ be a function with the property that for every $y \in \mathbb{R}$, the value of the expression

$$\sup_{x\in\mathbb{R}} [xy - f(x)]$$

is finite. Define $g(y) = \sup_{x\in\mathbb{R}} [xy - f(x)]$ for $y \in \mathbb{R}$. Then

(A) g is even if f is even.

(B) f must satisfy $\lim_{|x|\to\infty} \frac{f(x)}{|x|} = +\infty$.

(C) g is odd if f is even.

(D) f must satisfy $\lim_{|x|\to\infty} \frac{f(x)}{|x|} = -\infty$.

Question 10 (JAM 2021 • Q26 • ID: JAM_2021_Q26) MCQ +2 M Neg: −0.67

Q. 26 Let $f : [0, 1] \to [0, 1]$ be a non-constant continuous function such that $f \circ f = f$. Define

$$E_f = \{x \in [0, 1] : f(x) = x\}.$$

Then

(A) E_f is neither open nor closed.

(B) E_f is an interval.

(C) E_f is empty.

(D) E_f need not be an interval.

Question 11 (JAM 2021 • Q25 • ID: JAM_2021_Q25) **MCQ** **+2 M** **Neg: −0.67**

Q. 25 Let y be a twice differentiable function on $\mathbb{R}$ satisfying

$$y''(x) = 2 + e^{-|x|}, \quad x \in \mathbb{R},$$
$$y(0) = -1, \quad y'(0) = 0.$$

Then

(A) $y = 0$ has exactly one root.

(B) $y = 0$ has exactly two roots.

(C) $y = 0$ has more than two roots.

(D) there exists an $x_0 \in \mathbb{R}$ such that $y(x_0) \geq y(x)$ for all $x \in \mathbb{R}$.

Question 12 (JAM 2021 • Q21 • ID: JAM_2021_Q21) **MCQ** **+2 M** **Neg: −0.67**

Q. 21 Let $f : [0,1] \to [0,\infty)$ be a continuous function such that

$$\big(f(t)\big)^2 < 1 + 2\int_0^t f(s)\,ds, \text{ for all } t \in [0,1].$$

Then

(A) $f(t) < 1 + t$ for all $t \in [0,1]$.

(B) $f(t) > 1 + t$ for all $t \in [0,1]$.

(C) $f(t) = 1 + t$ for all $t \in [0,1]$.

(D) $f(t) < 1 + \frac{t}{2}$ for all $t \in [0,1]$.

Question 13 (JAM 2021 • Q18 • ID: JAM_2021_Q18) **MCQ** **+2 M** **Neg: −0.67**

Q. 18 Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \in (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}, \\ 1 - \frac{1}{p} & \text{if } x = \frac{n}{p},\ n \in \mathbb{Z} \setminus \{0\},\ p \in \mathbb{N} \text{ and } \gcd(n, p) = 1. \end{cases}$$

Then

(A) all $x \in \mathbb{Q} \setminus \{0\}$ are strict local minima for f.

(B) f is continuous at all $x \in \mathbb{Q}$.

(C) f is not continuous at all $x \in \mathbb{R} \setminus \mathbb{Q}$.

(D) f is not continuous at $x = 0$.

Question 14 (JAM 2021 • Q17 • ID: JAM_2021_Q17) **MCQ** **+2 M** **Neg: −0.67**

Q. 17 Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that for all $a, b \in \mathbb{R}$ with $a < b$,

$$\frac{f(b) - f(a)}{b - a} = f'\left(\frac{a+b}{2}\right).$$

Then

(A) f must be a polynomial of degree less than or equal to 2.

(B) f must be a polynomial of degree greater than 2.

(C) f is not a polynomial.

(D) f must be a linear polynomial.

Question 15 (JAM 2021 • Q7 • ID: JAM_2021_Q7) **MCQ** +1 M **Neg: −0.33**

JAM 2021 MATHEMATICS - MA

Q. 7 For every $n \in \mathbb{N}$, let $f_n : \mathbb{R} \to \mathbb{R}$ be a function. From the given choices, pick the statement that is the negation of

"For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$."

(A) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there does not exist any integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.

(B) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ for some integer $p > 0$.

(C) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$, there exists an integer $p > 0$ for which the inequality $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ holds.

(D) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$ and for every integer $p > 0$ the inequality $\sum_{i=1}^{p} |f_{N+i}(x)| \geq \epsilon$ holds.

Question 16 (JAM 2021 • Q4 • ID: JAM_2021_Q4) **MCQ** +1 M **Neg: −0.33**

JAM 2021 MATHEMATICS - MA

Q. 4 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that for all $x \in \mathbb{R}$,

$$\int_0^1 f(xt)\, dt = 0. \qquad (*)$$

Then

(A) f must be identically 0 on the whole of $\mathbb{R}$.

(B) there is an f satisfying $(*)$ that is identically 0 on $(0, 1)$ but not identically 0 on the whole of $\mathbb{R}$.

(C) there is an f satisfying $(*)$ that takes both positive and negative values.

(D) there is an f satisfying $(*)$ that is 0 at infinitely many points, but is not identically zero.

Question 17 (JAM 2021 • Q3 • ID: JAM_2021_Q3) **MCQ** **+1 M** **Neg: −0.33**

Q. 3 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f(x) = f(x+1)$ for all $x \in \mathbb{R}$. Then

(A) f is not necessarily bounded above.

(B) there exists a unique $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

(C) there is no $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

(D) there exist infinitely many $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.

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Question 18 (JAM 2021 • Q2 • ID: JAM_2021_Q2) **MCQ** **+1 M** **Neg: −0.33**

Q. 2 Let $P : \mathbb{R} \to \mathbb{R}$ be a continuous function such that $P(x) > 0$ for all $x \in \mathbb{R}$. Let y be a twice differentiable function on $\mathbb{R}$ satisfying $y''(x) + P(x)y'(x) - y(x) = 0$ for all $x \in \mathbb{R}$. Suppose that there exist two real numbers a, b $(a < b)$ such that $y(a) = y(b) = 0$. Then

(A) $y(x) = 0$ for all $x \in [a, b]$.

(B) $y(x) > 0$ for all $x \in (a, b)$.

(C) $y(x) < 0$ for all $x \in (a, b)$.

(D) $y(x)$ changes sign on (a, b).

Question 19 (JAM 2021 • Q1 • ID: JAM_2021_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let $0 < \alpha < 1$ be a real number. The number of differentiable functions $y : [0,1] \to [0,\infty)$, having continuous derivative on $[0,1]$ and satisfying

$$\begin{aligned} y'(t) &= (y(t))^{\alpha}, \quad t \in [0,1], \\ y(0) &= 0, \end{aligned}$$

is

(A) exactly one.

(B) exactly two.

(C) finite but more than two.

(D) infinite.

Question 20 (JAM 2020 • Q54 • ID: JAM_2020_Q54) NAT +2 M Neg: Nil (0)

Q. 54 Let $f(x) = \sqrt{x} + \alpha x,\ x > 0$ and

$$g(x) = a_0 + a_1(x-1) + a_2(x-1)^2$$

be the sum of the first three terms of the Taylor series of $f(x)$ around $x = 1$. If $g(3) = 3$, then α is ________.

Question 21 (JAM 2020 • Q44 • ID: JAM_2020_Q44) NAT +1 M Neg: Nil (0)

Q. 44 Let $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ and $f : S \to \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Then

$$\max\left\{ \delta : \left| x - \frac{1}{3} \right| < \delta \implies \left| f(x) - f\left(\frac{1}{3}\right) \right| < 1 \right\}$$

is ________. (rounded off to two decimal places)

Question 22 (JAM 2020 • Q43 • ID: JAM_2020_Q43) NAT +1 M Neg: Nil (0)

Q. 43 Let $f : \mathbb{R} \to \mathbb{R}$ be such that f, f', f'' are continuous functions with $f > 0$, $f' > 0$ and $f'' > 0$. Then

$$\lim_{x \to -\infty} \frac{f(x) + f'(x)}{2}$$

is ________.

Question 23 (JAM 2020 • Q33 • ID: JAM_2020_Q33) MSQ +2 M Neg: Nil (0)

Q. 33 Let $a, b \in \mathbb{R}$ and $a < b$. Which of the following statement(s) is/are true?

(A) There exists a continuous function $f : [a, b] \to (a, b)$ such that f is one-one

(B) There exists a continuous function $f : [a, b] \to (a, b)$ such that f is onto

(C) There exists a continuous function $f : (a, b) \to [a, b]$ such that f is one-one

(D) There exists a continuous function $f : (a, b) \to [a, b]$ such that f is onto

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Question 24 (JAM 2020 • Q31 • ID: JAM_2020_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 Let $a, b, c \in \mathbb{R}$ such that $a < b < c$. Which of the following is/are true for any continuous function $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(a) = b$, $f(b) = c$ and $f(c) = a$?

(A) There exists $\alpha \in (a, c)$ such that $f(\alpha) = \alpha$

(B) There exists $\beta \in (a, b)$ such that $f(\beta) = \beta$

(C) There exists $\gamma \in (a, b)$ such that $(f \circ f)(\gamma) = \gamma$

(D) There exists $\delta \in (a, c)$ such that $(f \circ f \circ f)(\delta) = \delta$

Question 25 (JAM 2020 • Q28 • ID: JAM_2020_Q28) **MCQ** **+2 M** **Neg: −0.67**

Q. 28 Let $f : [0,1] \to \mathbb{R}$ be a continuous function such that $f\left(\frac{1}{2}\right) = -\frac{1}{2}$ and

$$|f(x) - f(y) - (x - y)| \leq \sin\left(|x - y|^2\right)$$

for all $x, y \in [0,1]$. Then $\int_0^1 f(x)\,dx$ is

(A) $-\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Question 26 (JAM 2020 • Q21 • ID: JAM_2020_Q21) **MCQ** **+2 M** **Neg: −0.67**

JAM 2020 MATHEMATICS - MA

Q. 21 Consider the differential equation $L[y] = (y - y^2)dx + xdy = 0$. The function $f(x, y)$ is said to be an integrating factor of the equation if $f(x, y)L[y] = 0$ becomes exact.

If $f(x, y) = \frac{1}{x^2y^2}$, then

(A) f is an integrating factor and $y = 1 - kxy,\ k \in \mathbb{R}$ is NOT its general solution

(B) f is an integrating factor and $y = -1 + kxy,\ k \in \mathbb{R}$ is its general solution

(C) f is an integrating factor and $y = -1 + kxy,\ k \in \mathbb{R}$ is NOT its general solution

(D) f is NOT an integrating factor and $y = 1 + kxy,\ k \in \mathbb{R}$ is its general solution

Question 27 (JAM 2020 • Q19 • ID: JAM_2020_Q19) **MCQ** **+2 M** **Neg: −0.67**

Q. 19 Let M be a real 6×6 matrix. Let 2 and -1 be two eigenvalues of M. If $M^5 = aI + bM$, where $a, b \in \mathbb{R}$, then

(A) $a = 10, b = 11$ (B) $a = -11, b = 10$

(C) $a = -10, b = 11$ (D) $a = 10, b = -11$

Question 28 (JAM 2020 • Q11 • ID: JAM_2020_Q11) MCQ +2 M Neg: −0.67

Q. 11 Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $l = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$. Which of the following is true?

(A) If $l = 1$, then $\lim_{n\to\infty} a_n = 1$

(B) If $l = 1$, then $\lim_{n\to\infty} a_n = 0$

(C) If $l < 1$, then $\lim_{n\to\infty} a_n = 1$

(D) If $l < 1$, then $\lim_{n\to\infty} a_n = 0$

Question 29 (JAM 2020 • Q1 • ID: JAM_2020_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let $s_n = 1 + \frac{(-1)^n}{n}, n \in \mathbb{N}$. Then the sequence $\{s_n\}$ is

(A) monotonically increasing and is convergent to 1

(B) monotonically decreasing and is convergent to 1

(C) neither monotonically increasing nor monotonically decreasing but is convergent to 1

(D) divergent

Question 30 (JAM 2019 • Q47 • ID: JAM_2019_Q47) NAT +1 M Neg: Nil (0)

Q.47 If

$$g(x) = \int_{x(x-2)}^{4x-5} f(t)\,dt\,, \text{ where } f(x) = \sqrt{1+3x^4} \text{ for } x \in \mathbb{R}$$

then $g'(1) =$ _______

Question 31 (JAM 2019 • Q46 • ID: JAM_2019_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let $f:[0,1] \to \mathbb{R}$ be given by

$$f(x) = \frac{\left(1+x^{\frac{1}{3}}\right)^3 + \left(1-x^{\frac{1}{3}}\right)^3}{8(1+x)}.$$

Then

$$\max\{f(x): x \in [0,1]\} - \min\{f(x): x \in [0,1]\}$$

is ____________

Question 32 (JAM 2019 • Q45 • ID: JAM_2019_Q45) NAT +1 M Neg: Nil (0)

Q.45 The coefficient of $\left(x - \frac{\pi}{2}\right)$ in the Taylor series expansion of the function

$$f(x) = \begin{cases} \dfrac{4(1-\sin x)}{2x-\pi}, & x \neq \dfrac{\pi}{2} \\ 0, & x = \dfrac{\pi}{2} \end{cases}$$

about $x = \frac{\pi}{2}$, is ____________

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Question 33 (JAM 2019 • Q38 • ID: JAM_2019_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Let $f\colon \left(0, \frac{\pi}{2}\right) \to \mathbb{R}$ be given by

$$f(x) = (\sin x)^{\pi} - \pi \sin x + \pi.$$

Then which of the following statements is/are TRUE?

(A) f is an increasing function
(B) f is a decreasing function
(C) $f(x) > 0$ for all $x \in \left(0, \frac{\pi}{2}\right)$
(D) $f(x) < 0$ for some $x \in \left(0, \frac{\pi}{2}\right)$

Question 34 (JAM 2019 • Q37 • ID: JAM_2019_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let

$$f(x) = \cos(|\pi - x|) + (x - \pi)\sin|x| \text{ and } g(x) = x^2 \text{ for } x \in \mathbb{R}.$$

If $h(x) = f\big(g(x)\big)$, then

(A) h is not differentiable at $x = 0$
(B) $h'\left(\sqrt{\pi}\right) = 0$
(C) $h''(x) = 0$ has a solution in $(-\pi, \pi)$
(D) there exists $x_0 \in (-\pi, \pi)$ such that $h(x_0) = x_0$

Question 35 (JAM 2019 • Q28 • ID: JAM_2019_Q28) MCQ +2 M Neg: −0.67

Q.28 Let $\{a_n\}$ be a sequence of positive real numbers. The series $\sum_{n=1}^{\infty} a_n$ converges if the series

(A) $\sum_{n=1}^{\infty} a_n^2$ converges
(B) $\sum_{n=1}^{\infty} \frac{a_n}{2^n}$ converges
(C) $\sum_{n=1}^{\infty} \frac{a_{n+1}}{a_n}$ converges
(D) $\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}}$ converges

Question 36 (JAM 2019 • Q21 • ID: JAM_2019_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $x,\ x+e^x$ and $1+x+e^x$ be solutions of a linear second order ordinary differential equation with constant coefficients. If $y(x)$ is the solution of the same equation satisfying $y(0)=3$ and $y'(0)=4$, then $y(1)$ is equal to

(A) $e+1$ (B) $2e+3$ (C) $3e+2$ (D) $3e+1$

Question 37 (JAM 2019 • Q18 • ID: JAM_2019_Q18) MCQ +2 M Neg: −0.67

Q.18 For which one of the following values of k, the equation

$$2x^3+3x^2-12x-k=0$$

has three distinct real roots?

(A) 16 (B) 20 (C) 26 (D) 31

Question 38 (JAM 2019 • Q11 • ID: JAM_2019_Q11) MCQ +2 M Neg: −0.67

Q.11 Let H and K be subgroups of $\mathbb{Z}_{144}$. If the order of H is 24 and the order of K is 36, then the order of the subgroup $H\cap K$ is

(A) 3 (B) 4 (C) 6 (D) 12

Question 39 (JAM 2019 • Q7 • ID: JAM_2019_Q7) MCQ +1 M Neg: −0.33

Q.7 If $y(x)=\lambda e^{2x}+e^{\beta x}$, $\beta\neq 2$, is a solution of the differential equation

$$\frac{d^2y}{dx^2}+\frac{dy}{dx}-6y=0$$

satisfying $\frac{dy}{dx}(0)=5$, then $y(0)$ is equal to

(A) 1 (B) 4 (C) 5 (D) 9

Question 40 (JAM 2018 • Q55 • ID: JAM_2018_Q55) NAT +2 M Neg: Nil (0)

Q.55 If $\alpha = \int_{\pi/6}^{\pi/3} \frac{\sin t + \cos t}{\sqrt{\sin 2t}}\, dt$, then the value of $\left(2 \sin\frac{\alpha}{2} + 1\right)^2$ is ______________

Question 41 (JAM 2018 • Q52 • ID: JAM_2018_Q52) NAT +2 M Neg: Nil (0)

Q.52 Let $f\colon \mathbb{R} \to \mathbb{R}$ be such that f'' is continuous on $\mathbb{R}$ and $f(0) = 1,\ f'(0) = 0$ and $f''(0) = -1$.

Then $\lim_{x\to\infty} \left(f\left(\sqrt{\frac{2}{x}}\right)\right)^x$ is ________________ (correct up to three decimal places).

Question 42 (JAM 2018 • Q31 • ID: JAM_2018_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let $f\colon \mathbb{R}\setminus\{0\} \to \mathbb{R}$ be defined by $f(x) = x + \frac{1}{x^3}$. On which of the following interval(s) is f one-one?

(A) $(-\infty, -1)$ (B) $(0, 1)$ (C) $(0, 2)$ (D) $(0,\ \infty)$

Question 43 (JAM 2018 • Q30 • ID: JAM_2018_Q30) MCQ +2 M Neg: −0.67

Q.30 For $x > \frac{-1}{2}$, let $f_1(x) = \frac{2x}{1+2x}$, $f_2(x) = \log_e(1+2x)$ and $f_3(x) = 2x$. Then which one of the following is TRUE?

(A) $f_3(x) < f_2(x) < f_1(x)$ for $0 < x < \frac{\sqrt{3}}{2}$

(B) $f_1(x) < f_3(x) < f_2(x)$ for $x > 0$

(C) $f_1(x) + f_2(x) < \frac{f_3(x)}{2}$ for $x > \frac{\sqrt{3}}{2}$

(D) $f_2(x) < f_1(x) < f_3(x)$ for $x > 0$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each

Question 44 (JAM 2018 • Q21 • ID: JAM_2018_Q21) MCQ +2 M Neg: −0.67

Q.21 Let U, V and W be finite dimensional real vector spaces, $T: U \to V$, $S: V \to W$ and $P: W \to U$ be linear transformations. If range (ST) = nullspace (P), nullspace (ST) = range (P) and rank (T) = rank (S), then which one of the following is TRUE?

(A) nullity of T = nullity of S

(B) dimension of $U \neq$ dimension of W

(C) If dimension of $V = 3$, dimension of $U = 4$, then P is not identically zero

(D) If dimension of $V = 4$, dimension of $U = 3$ and T is one-one, then P is identically zero

Question 45 (JAM 2018 • Q11 • ID: JAM_2018_Q11)   MCQ   +2 M   Neg: −0.67

Q. 11 – Q. 30 carry two marks each.

Q.11 Let $a_n = \begin{cases} 2 + \frac{(-1)^{\frac{n-1}{2}}}{n}, & \text{if } n \text{ is odd} \\ 1 + \frac{1}{2^n}, & \text{if } n \text{ is even} \end{cases}, \; n \in \mathbb{N}.$

Then which one of the following is TRUE?

(A) $\sup\{a_n \mid n \in \mathbb{N}\} = 3$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(B) $\liminf\,(a_n) = \limsup\,(a_n) = \frac{3}{2}$

(C) $\sup\{a_n \mid n \in \mathbb{N}\} = 2$ and $\inf\{a_n \mid n \in \mathbb{N}\} = 1$

(D) $\liminf\,(a_n) = 1$ and $\limsup\,(a_n) = 3$

Question 46 (JAM 2018 • Q1 • ID: JAM_2018_Q1)   MCQ   +1 M   Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Which one of the following is TRUE?

(A) $\mathbb{Z}_n$ is cyclic if and only if n is prime

(B) Every proper subgroup of $\mathbb{Z}_n$ is cyclic

(C) Every proper subgroup of S_4 is cyclic

(D) If every proper subgroup of a group is cyclic, then the group is cyclic

Question 47 (JAM 2017 • Q57 • ID: JAM_2017_Q57)   NAT   +2 M   Neg: Nil (0)

Q.57 For $x > 1$, let

$$f(x) = \int_1^x \left(\sqrt{\log t} - \frac{1}{2}\log\sqrt{t}\right) dt$$

The number of tangents to the curve $y = f(x)$ parallel to the line $x + y = 0$ is ____________

Question 48 (JAM 2017 • Q53 • ID: JAM_2017_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $f(x) = \frac{\sin \pi x}{\pi \sin x}$, $x \in (0, \pi)$, and let $x_0 \in (0, \pi)$ be such that $f'(x_0) = 0$. Then

$$\left(f(x_0)\right)^2 (1 + (\pi^2 - 1)\sin^2 x_0) = \_\_\_\_\_\_\_\_$$

Question 49 (JAM 2017 • Q49 • ID: JAM_2017_Q49) NAT +1 M Neg: Nil (0)

Q.49 For $x > 0$, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x. Then

$$\lim_{x \to 0^+} x\left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{10}{x} \right\rfloor \right) = \_\_\_\_\_\_\_\_$$

Question 50 (JAM 2017 • Q45 • ID: JAM_2017_Q45) NAT +1 M Neg: Nil (0)

Q.45

$$\left(\int_0^1 x^4 (1 - x)^5 \, dx \right)^{-1} = \_\_\_\_\_\_\_\_$$

Question 51 (JAM 2017 • Q25 • ID: JAM_2017_Q25) MCQ +2 M Neg: −0.67

Q.25

$$\sum_{n=1}^{\infty} \tan^{-1} \frac{2}{n^2} =$$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) π

Question 52 (JAM 2017 • Q20 • ID: JAM_2017_Q20) MCQ +2 M Neg: −0.67

Q.20 Let $f: \mathbb{R} \to [0, \infty)$ be a continuous function. Then which one of the following is NOT TRUE?

(A) There exists $x \in \mathbb{R}$ such that $f(x) = \frac{f(0)+f(1)}{2}$

(B) There exists $x \in \mathbb{R}$ such that $f(x) = \sqrt{f(-1)f(1)}$

(C) There exists $x \in \mathbb{R}$ such that $f(x) = \int_{-1}^{1} f(t)dt$

(D) There exists $x \in \mathbb{R}$ such that $f(x) = \int_{0}^{1} f(t)dt$

Question 53 (JAM 2017 • Q19 • ID: JAM_2017_Q19) MCQ +2 M Neg: −0.67

Q.19 Let $f: \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f(2) = 2$ and

$$|f(x) - f(y)| \leq 5(|x-y|)^{3/2}$$

for all $x \in \mathbb{R}, y \in \mathbb{R}$. Let $g(x) = x^3 f(x)$. Then $g'(2) =$

(A) 5 (B) $\frac{15}{2}$ (C) 12 (D) 24

Question 54 (JAM 2017 • Q10 • ID: JAM_2017_Q10) MCQ +1 M Neg: −0.33

Q.10 If

$$f(x) = \begin{cases} 1+x & \text{if } x < 0 \\ (1-x)(px+q) & \text{if } x \geq 0 \end{cases}$$

satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (p, q) is

(A) $(2, -1)$ (B) $(-2, -1)$ (C) $(-2, 1)$ (D) $(2, 1)$

Question 55 (JAM 2017 • Q8 • ID: JAM_2017_Q8) **MCQ** **+1 M** **Neg: −0.33**

Q.8 Let

$$f(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right), \qquad x \neq 0.$$

Write $L = \lim_{x \to 0^-} f(x)$ and $R = \lim_{x \to 0^+} f(x)$. Then which one of the following is TRUE?

(A) L exists but R does not exist

(B) L does not exist but R exists

(C) Both L and R exist

(D) Neither L nor R exists

Question 56 (JAM 2017 • Q7 • ID: JAM_2017_Q7) **MCQ** **+1 M** **Neg: −0.33**

Q.7 Let $f_1(x), f_2(x), g_1(x), g_2(x)$ be differentiable functions on $\mathbb{R}$. Let $F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$ be the determinant of the matrix $\begin{bmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{bmatrix}$. Then $F'(x)$ is equal to

(A) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2'(x) & g_2(x) \end{vmatrix}$

(B) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(C) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} - \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(D) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$

Question 57 (JAM 2017 • Q2 • ID: JAM_2017_Q2) **MCQ** **+1 M** **Neg: −0.33**

Q.2 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be a differentiable function such that φ' is strictly increasing with $\varphi'(1) = 0$. Let α and β denote the minimum and maximum values of $\varphi(x)$ on the interval $[2, 3]$, respectively. Then which one of the following is TRUE?

(A) $\beta = \varphi(3)$ (B) $\alpha = \varphi(2.5)$ (C) $\beta = \varphi(2.5)$ (D) $\alpha = \varphi(3)$

Question 58 (JAM 2016 • Q49 • ID: JAM_2016_Q49) NAT +1 M Neg: Nil (0)

Q.49 Let $f\colon (0,\infty) \to \mathbb{R}$ be a continuous function such that

$$\int_0^x f(t)dt = -2 + \frac{x^2}{2} + 4x\sin 2x + 2\cos 2x.$$

Then the value of $\frac{1}{\pi} f\left(\frac{\pi}{4}\right)$ is ____________

Question 59 (JAM 2016 • Q46 • ID: JAM_2016_Q46) NAT +1 M Neg: Nil (0)

Q.46 If $f\colon (-1,\infty) \to \mathbb{R}$ defined by $f(x) = \frac{x}{1+x}$ is expressed as

$$f(x) = \frac{2}{3} + \frac{1}{9}(x-2) + \frac{c(x-2)^2}{(1+\xi)^3},$$

where ξ lies between 2 and x, then the value of c is ____________

Question 60 (JAM 2016 • Q37 • ID: JAM_2016_Q37) MSQ +2 M Neg: Nil (0)

Q.37 Let $P(x) = \left(\frac{5}{13}\right)^x + \left(\frac{12}{13}\right)^x - 1$ for all $x \in \mathbb{R}$. Then which of the following statement(s) is(are) TRUE?

(A) The equation $P(x) = 0$ has exactly one solution in $\mathbb{R}$
(B) $P(x)$ is strictly increasing for all $x \in \mathbb{R}$
(C) The equation $P(x) = 0$ has exactly two solutions in $\mathbb{R}$
(D) $P(x)$ is strictly decreasing for all $x \in \mathbb{R}$

MA 9/13

Question 61 (JAM 2016 • Q32 • ID: JAM_2016_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The value(s) of the integral

$$\int_{-\pi}^{\pi} |x| \cos nx \, dx \ , \ \ n \geq 1$$

is (are)

(A) 0 when n is even
(B) 0 when n is odd
(C) $-\frac{4}{n^2}$ when n is even
(D) $-\frac{4}{n^2}$ when n is odd

MA 8/13

Question 62 (JAM 2016 • Q31 • ID: JAM_2016_Q31) MSQ +2 M Neg: Nil (0)

Q.31 Let $\{s_n\}$ be a sequence of positive real numbers satisfying

$$2\, s_{n+1} = s_n^2 + \frac{3}{4}, \qquad n \geq 1.$$

If α and β are the roots of the equation $x^2 - 2x + \frac{3}{4} = 0$ and $\alpha < s_1 < \beta$, then which of the following statement(s) is(are) TRUE ?

(A) $\{s_n\}$ is monotonically decreasing
(B) $\{s_n\}$ is monotonically increasing
(C) $\lim_{n\to\infty} s_n = \alpha$
(D) $\lim_{n\to\infty} s_n = \beta$

Question 63 (JAM 2016 • Q21 • ID: JAM_2016_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $y(x)$ be the solution of the differential equation

$$\frac{d}{dx}\left(x\frac{dy}{dx}\right) = x; \quad y(1) = 0, \quad \left.\frac{dy}{dx}\right|_{x=1} = 0\,.$$

Then $y(2)$ is

(A) $\frac{3}{4} + \frac{1}{2}\ln 2$

(B) $\frac{3}{4} - \frac{1}{2}\ln 2$

(C) $\frac{3}{4} + \ln 2$

(D) $\frac{3}{4} - \ln 2$

Question 64 (JAM 2016 • Q11 • ID: JAM_2016_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S be the series

$$\sum_{k=1}^{\infty} \frac{1}{(2k-1)\, 2^{(2k-1)}}$$

and T be the series

$$\sum_{k=2}^{\infty} \left(\frac{3k-4}{3k+2}\right)^{\frac{(k+1)}{3}}$$

of real numbers. Then which one of the following is TRUE?

(A) Both the series S and T are convergent
(B) S is convergent and T is divergent
(C) S is divergent and T is convergent
(D) Both the series S and T are divergent

MA 4/13

Question 65 (JAM 2016 • Q5 • ID: JAM_2016_Q5) MCQ +1 M Neg: −0.33

Q.5 Let f be a strictly monotonic continuous real valued function defined on $[a, b]$ such that $f(a) < a$ and $f(b) > b$. Then which one of the following is TRUE?

(A) There exists exactly one $c \in (a, b)$ such that $f(c) = c$
(B) There exist exactly two points $c_1, c_2 \in (a, b)$ such that $f(c_i) = c_i,\ i = 1,2$
(C) There exists no $c \in (a, b)$ such that $f(c) = c$
(D) There exist infinitely many points $c \in (a, b)$ such that $f(c) = c$

MA 3/13

Question 66 (JAM 2016 • Q3 • ID: JAM_2016_Q3) MCQ +1 M Neg: −0.33

Q.3 Let $f : [-1, 1] \to \mathbb{R}$ be a continuous function. Then the integral

$$\int_0^{\pi} x\, f(\sin x)\ dx$$

is equivalent to

(A) $$\frac{\pi}{2}\int_0^{\pi} f(\sin x)\ dx$$

(B) $$\frac{\pi}{2}\int_0^{\pi} f(\cos x)\ dx$$

(C) $$\pi\int_0^{\pi} f(\cos\ x)\ dx$$

(D) $$\pi\int_0^{\pi} f(\sin x)\ dx$$

Question 67 (JAM 2015 • Q59 • ID: JAM_2015_Q59) NAT +2 M Neg: Nil (0)

Q.19 The limit

$$\lim_{x \to 0+} \frac{9}{x}\left(\frac{1}{\tan^{-1} x} - \frac{1}{x}\right)$$

is equal to ______________

Question 68 (JAM 2015 • Q56 • ID: JAM_2015_Q56) NAT +2 M Neg: Nil (0)

Q.16

The coefficient of $\left(x - \frac{\pi}{4}\right)^3$ in the Taylor series expansion of the function

$$f(x) = 3 \sin x \cos\left(x + \frac{\pi}{4}\right), \qquad x \in \mathbb{R}$$

about the point $\frac{\pi}{4}$, correct upto three decimal places, is ______________

Question 69 (JAM 2015 • Q50 • ID: JAM_2015_Q50) NAT +1 M Neg: Nil (0)

Q.10

Let $f: (0,1) \to \mathbb{R}$ be a continuously differentiable function such that f' has finitely many zeros in $(0,1)$ and f' changes sign at exactly two of these points. Then for any $y \in \mathbb{R}$, the maximum number of solutions to $f(x) = y$ in $(0,1)$ is ______________

Q. 11 – Q. 20 carry two marks each

Question 70 (JAM 2015 • Q49 • ID: JAM_2015_Q49) NAT +1 M Neg: Nil (0)

Q.9

Let $f: \mathbb{R} \longrightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^6 - 1, & x \in \mathbb{Q} \\ 1 - x^6, & x \notin \mathbb{Q} \end{cases}$$

The number of points at which f is continuous, is ______________

Question 71 (JAM 2015 • Q40 • ID: JAM_2015_Q40) MSQ +2 M Neg: Nil (0)

Q.10

Let $f: (-1,1) \longrightarrow \mathbb{R}$ be the function defined by

$$f(x) = x^2 e^{1/(1-x^2)}$$

Then

(A) f is decreasing in $(-1,0)$

(B) f is increasing in $(0,1)$

(C) $f(x) = 1$ has two solutions in $(-1,1)$

(D) $f(x) = 1$ has no solutions in $(-1,1)$

MA 13/16

Question 72 (JAM 2015 • Q39 • ID: JAM_2015_Q39) MSQ +2 M Neg: Nil (0)

Q.9

Let $f, g : [0,1] \to [0,1]$ be functions. Let $R(f)$ and $R(g)$ be the ranges of f and g, respectively. Which of the following statements is (are) true?

(A) If $f(x) \leq g(x)$ for all $x \in [0,1]$, then $\sup R(f) \leq \inf R(g)$

(B) If $f(x) \leq g(x)$ for some $x \in [0,1]$, then $\inf R(f) \leq \sup R(g)$

(C) If $f(x) \leq g(y)$ for some $x, y \in [0,1]$, then $\inf R(f) \leq \sup R(g)$

(D) If $f(x) \leq g(y)$ for all $x, y \in [0,1]$, then $\sup R(f) \leq \inf R(g)$

Question 73 (JAM 2015 • Q37 • ID: JAM_2015_Q37) MSQ +2 M Neg: Nil (0)

Q.7 Which of the following statements is (are) true on the interval $\left(0, \frac{\pi}{2}\right)$?

(A) $\cos x < \cos(\sin x)$

(B) $\tan x < x$

(C) $\sqrt{1+x} < 1 + \frac{x}{2} - \frac{x^2}{8}$

(D) $\frac{1-x^2}{2} < \ln(2+x)$

Question 74 (JAM 2015 • Q31 • ID: JAM_2015_Q31) MSQ +2 M Neg: Nil (0)

Q.1

Let $f: \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \int_{-5}^{x} (t-1)^3 dt$.
In which of the following interval(s), f takes the value 1?

(A) $[-6,\ 0]$ (B) $[-2,\ 4]$ (C) $[2,\ 8]$ (D) $[6,\ 12]$

Question 75 (JAM 2015 • Q28 • ID: JAM_2015_Q28) MCQ +2 M Neg: −0.67

Q.28

For $m, n \in \mathbb{N}$, define $f_{m,n}(x) = \begin{cases} x^m \sin\left(\frac{1}{x^n}\right), & x \neq 0 \\ 0 & x = 0 \end{cases}$

Then at $x = 0$, $f_{m,n}$ is

(A) differentiable for each pair m, n with $m > n$

(B) differentiable for each pair m, n with $m < n$

(C) not differentiable for each pair m, n with $m > n$

(D) not differentiable for each pair m, n with $m < n$

Question 76 (JAM 2015 • Q27 • ID: JAM_2015_Q27) MCQ +2 M Neg: −0.67

Q.27

For $n \geq 2$, let $f_n: \mathbb{R} \to \mathbb{R}$ be given by $f_n(x) = x^n \sin x$. Then at $x = 0$, f_n has a

(A) local maximum if n is even

(B) local maximum if n is odd

(C) local minimum if n is even

(D) local minimum if n is odd

Question 77 (JAM 2015 • Q25 • ID: JAM_2015_Q25) MCQ +2 M Neg: −0.67

Q.25

For what real values of x and y, does the integral $\int_x^y (6 - t - t^2)dt$ attain its maximum?

(A) $x = -3,\ y = 2$

(B) $x = 2,\ y = 3$

(C) $x = -2,\ y = 2$

(D) $x = -3,\ y = 4$

Question 78 (JAM 2015 • Q21 • ID: JAM_2015_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $f\colon \mathbb{R} \to \mathbb{R}$ be a strictly increasing continuous function. If $\{a_n\}$ is a sequence in $[0,1]$, then the sequence $\{f(a_n)\}$ is

(A) increasing
(B) bounded
(C) convergent
(D) not necessarily bounded

Question 79 (JAM 2015 • Q11 • ID: JAM_2015_Q11) MCQ +2 M Neg: −0.67

Q.11 Let S_3 be the group of permutations of three distinct symbols. The direct sum $S_3 \oplus S_3$ has an element of order

(A) 4
(B) 6
(C) 9
(D) 18

Question 80 (JAM 2015 • Q1 • ID: JAM_2015_Q1) MCQ +1 M Neg: −0.33

Q.1 Suppose N is a normal subgroup of a group G. Which one of the following is true?

(A) If G is an infinite group then G/N is an infinite group
(B) If G is a nonabelian group then G/N is a nonabelian group
(C) If G is a cyclic group then G/N is an abelian group
(D) If G is an abelian group then G/N is a cyclic group

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of One Real Variable (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**0**
2	NAT	+2	**15**
3	NAT	+2	**0**
4	NAT	+2	**10.8 to 10.9**
5	NAT	+1	**2**
6	MSQ	+2	**B, C, D**
7	MSQ	+2	**A, C**
8	MSQ	+2	**A, B**
9	MSQ	+2	**A, B**
10	MCQ	+2	**B**
11	MCQ	+2	**B**
12	MCQ	+2	**A**
13	MCQ	+2	**A**
14	MCQ	+2	**A**
15	MCQ	+1	**C**
16	MCQ	+1	**A**
17	MCQ	+1	**D**
18	MCQ	+1	**A**
19	MCQ	+1	**D**
20	NAT	+2	**0.5 to 0.5**
21	NAT	+1	**0.08 to 0.09**
22	NAT	+1	**0 to INF**
23	MSQ	+2	**A;C;D**
24	MSQ	+2	**A;C;D**
25	MCQ	+2	**A**
26	MCQ	+2	**C**
27	MCQ	+2	**A**

Q. No.	Type	Marks	Official Key
28	MCQ	+2	**D**
29	MCQ	+1	**C**
30	NAT	+1	**8 to 8**
31	NAT	+1	**0.25 to 0.25**
32	NAT	+1	**1 to 1**
33	MSQ	+2	**B, C**
34	MSQ	+2	**B, C, D**
35	MCQ	+2	**C**
36	MCQ	+2	**D**
37	MCQ	+2	**A**
38	MCQ	+2	**D**
39	MCQ	+1	**C**
40	NAT	+2	**2.9 to 3.1**
41	NAT	+2	**0.350 to 0.380**
42	MSQ	+2	**B**
43	MCQ	+2	**Marks to All**
44	MCQ	+2	**C**
45	MCQ	+2	**A**
46	MCQ	+1	**B**
47	NAT	+2	**0.9 to 1.1**
48	NAT	+2	**0.9 to 1.1**
49	NAT	+1	**54.9 to 55.1**
50	NAT	+1	**1259.9 to 1260.1**
51	MCQ	+2	**C**
52	MCQ	+2	**C**
53	MCQ	+2	**D**
54	MCQ	+1	**D**

Q. No.	Type	Marks	Official Key
55	MCQ	+1	**A**
56	MCQ	+1	**B**
57	MCQ	+1	**A**
58	NAT	+1	**0.25 to 0.25**
59	NAT	+1	**−1 to −1**
60	MSQ	+2	**A;D**
61	MSQ	+2	**A;D**
62	MSQ	+2	**A;C**
63	MCQ	+2	**B**
64	MCQ	+2	**B**
65	MCQ	+1	**A**
66	MCQ	+1	**A**
67	NAT	+2	**3**
68	NAT	+2	**1.41 to 1.42**
69	NAT	+1	**3**
70	NAT	+1	**2**
71	MSQ	+2	**A;B;C**
72	MSQ	+2	**B;C;D**
73	MSQ	+2	**A;D**
74	MSQ	+2	**A;C;D**
75	MCQ	+2	**A**
76	MCQ	+2	**D**
77	MCQ	+2	**A**
78	MCQ	+2	**B**
79	MCQ	+2	**B**
80	MCQ	+1	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.

- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.